Memorization and Generalization under Training-Label Randomization on Tabular Data

Tianying Chen, Linjie Yang, and Elsey Chen
Department of Computer Science, University of California, Davis
ECS 171, Group 8
{tiaychen, yljyang, elschen}@ucdavis.edu

***Abstract*—A model that predicts its training labels well may have learned a transferable rule, or it may simply have memorized the answers. Zhang et al. showed that deep networks can fit completely random image labels, so training error alone cannot explain generalization. We carry their randomization test over to tabular data. On the UCI Dry Bean dataset (13,543 unique observations, 16 numerical features, seven classes), we train logistic regression (LR), a random forest (RF), and a multilayer perceptron (MLP) while replacing each training label with a uniformly random class with probability $p \in \{0, 0.25, 0.5, 0.75, 1\}$, using three paired seeds and clean validation and test labels. With clean labels, test Macro-F1 is 0.9404, 0.9361, and 0.9346 for LR, RF, and MLP. With fully random labels, RF still reproduces every training label, yet its clean-test Macro-F1 drops to 0.1314; its out-of-bag accuracy exposes this collapse without touching the test set. The MLP fits only 38% of random labels within its 500-epoch budget, but in an exploratory capacity study an unregularized 512–256 network fits 97% of random labels on a 3,000-row subset while still generalizing well when trained on clean labels. Under heavy but partial noise the models diverge: at $p = 0.75$, LR retains 0.9073 Macro-F1, compared with 0.6440 for RF and 0.5856 for MLP. The code, recorded results, and an online inference demo are publicly available.**

***Index Terms*—label noise, memorization, generalization, tabular classification, reproducibility**

I. Introduction

A supervised classifier learns from input–label pairs, but a high training score does not tell us what it has learned. A flexible model can associate each training input with an arbitrary answer instead of recovering a relationship that also holds for new inputs. Label randomization makes the two possibilities observable: inputs stay the same, some training answers are replaced, and evaluation still asks for the original categories on unseen examples. Fitting the altered answers and predicting the true categories then become separate measurements.

Zhang et al. [1] used this idea to show that standard image-classification networks reach zero training error on random labels, even with explicit regularization, while test accuracy falls to chance. The result challenged explanations of generalization based only on model family, explicit regularization, or the ability to fit training data. Their experiments used convolutional networks on CIFAR-10 and ImageNet. Whether the same separation appears for small tabular datasets and classical learners is a natural follow-up question.

We study seven-class classification of dry beans from 16 shape and size measurements that were extracted from images [2], [3]. Our models receive these numbers, not pixels. We compare a linear classifier, a tree ensemble, and a feedforward network under a single controlled protocol, summarized in Fig. 1. Our two research questions are:

- **RQ1:** How does training-label randomization affect clean-test Macro-F1 for LR, RF, and an MLP?
- **RQ2:** Does stronger agreement with randomized training labels go together with better prediction of the original labels on unseen observations?

Hyperparameters are chosen once on clean labels and then held fixed, so the experiment measures how well-tuned clean-data models react to increasingly uninformative supervision; it does not search for the most noise-robust configuration of each family.

Our contributions are: (i) a paired randomization experiment that distinguishes the probability of redrawing a label from the fraction that actually changes; (ii) reporting training accuracy against both the supplied and the original labels next to clean-test metrics, together with RF out-of-bag estimates, so that fitting capacity is not confused with generalization; and (iii) an exploratory capacity study that connects the MLP result back to the original paper. All results come with saved predictions, a reproducible pipeline, and an online demo.

II. Background and Related Work

A. Random Labels as a Diagnostic

Zhang et al. [1] train networks on CIFAR-10 and ImageNet after corrupting the data in several ways: replacing a fraction p of labels with uniformly random classes, shuffling pixels, or replacing images with Gaussian noise. Networks fit all of these to zero training error; training slows down as corruption grows, and test error rises smoothly toward chance. They also show that weight decay, dropout, and data augmentation do not prevent random-label fitting, and prove that a two-layer ReLU network with $2n + d$ parameters can represent any labeling of n points in d dimensions. A later retrospective [4] restates these findings and reviews subsequent theory. Table I maps the original experiments to ours.

Arpit et al. [5] refine the picture: networks fit real patterns before memorizing noise, and regularization slows memorization more than learning. Rolnick et al. [6] report that deep networks still reach high test accuracy when clean examples are vastly outnumbered by uniformly mislabeled ones, provided enough clean data remain. These results motivate our

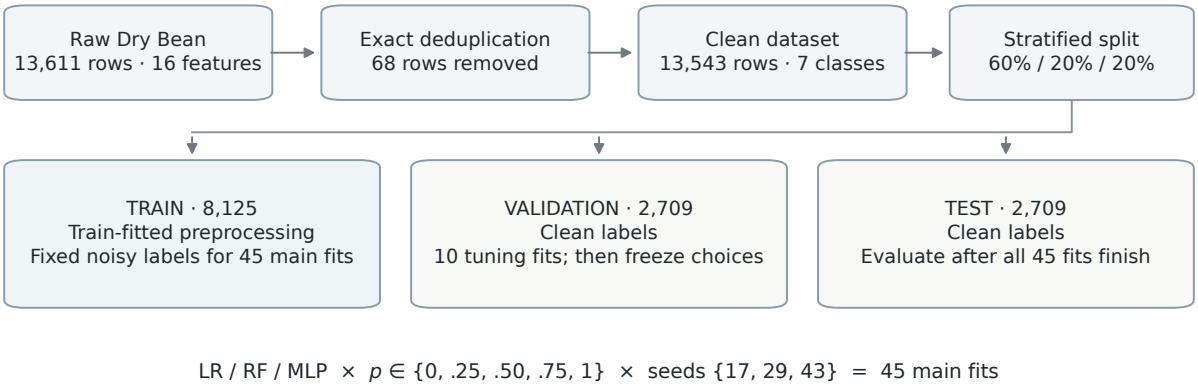


Fig. 1. Study design. Exact duplicates are removed before a fixed stratified split. Only training labels are randomized; validation labels are used for the ten clean-label tuning fits, and the clean test set is evaluated once after all 45 main fits are complete.

TABLE I
EXPERIMENTS OF ZHANG ET AL. [1] AND THEIR COUNTERPARTS IN THIS STUDY.

Original experiment	This study
Label corruption level p vs. test error	Five levels of p, three model families (Table IV, Fig. 4)
Random labels reach zero training error	RF: yes; MLP: partially at 500 epochs, almost fully with more capacity and epochs (Sec. V-E)
Training loss curves by corruption type	MLP loss curves by p (Fig. 7)
Explicit regularization ablation	Clean-label α grid; $\alpha = 0$ in the capacity study
Shuffled / Gaussian inputs	Not run
Finite-sample expressivity theorem	Not studied

TABLE II
CLASS COUNTS AFTER DEDUPLICATION AND IN THE FIXED SPLIT.

Class	Cleaned	Train	Val.	Test
BARBUNYA	1,322	793	264	265
BOMBAY	522	313	105	104
CALI	1,630	978	326	326
DERMASON	3,546	2,127	710	709
HOROZ	1,860	1,116	372	372
SEKER	2,027	1,216	405	406
SIRA	2,636	1,582	527	527
Total	13,543	8,125	2,709	2,709

recording of MLP training curves and of the gap between noisy- and clean-label training accuracy.

B. Label Noise and Model Families

Learning from labels flipped uniformly at random is a classical problem [7]. Under such symmetric noise, the Bayes-optimal decision rule is unchanged, and several losses are provably robust to it [8], [9]; Frénay and Verleysen [10] survey the broader literature. We do not propose a noise-correction method. Instead, we measure how unmodified estimators respond.

Random forests average randomized decision trees [11]. Fully grown trees interpolate their training data, and Wyner et al. [12] argue that such interpolating ensembles can still generalize because averaging localizes the effect of noisy points. Tree-based models are also strong baselines on tabular data [13]. LR provides a regularized linear rule after standardization, and the MLP supplies learned nonlinear features trained with Adam [14]. All estimators use scikit-learn [15].

Dry Bean was introduced by Koklu and Ozkan [2], who report cross-validated accuracy on the full dataset. Because we deduplicate, use a fixed split, and report Macro-F1, their numbers are not directly comparable to ours.

III. DATASET AND EXPLORATORY ANALYSIS

A. Source, Audit, and Split

The UCI release contains 13,611 observations, 16 numerical predictors, and seven bean classes [3]. The audit found no missing or nonfinite values, no feature-identical rows with conflicting labels, and 68 exact duplicate rows, all in HOROZ. Keeping the first occurrence leaves 13,543 distinct rows. Deduplicating before splitting prevents identical observations from appearing in both training and evaluation data; it reduces HOROZ from 1,928 to 1,860.

We first reserve 20% of the cleaned data for testing (stratified, random state 171) and then 25% of the remainder for validation (random state 172). This gives 8,125 training, 2,709 validation, and 2,709 test observations (Table II). Every experimental condition uses this split.

B. Feature Structure and Class Imbalance

Exploratory analysis uses only the training partition. DERMASON, the largest class, has 2,127 training observations and BOMBAY has 313, a ratio of 6.8 (Fig. 2). Because accuracy would be dominated by the large classes, we use Macro-F1

as the primary metric and also report balanced accuracy and per-class results.

The predictors describe related aspects of geometry. Area, ConvexArea, Perimeter, and EquivDiameter measure size; the major and minor axis lengths measure extent; aspect ratio, eccentricity, compactness, roundness, solidity, extent, and four shape factors describe normalized shape. We keep the release's spelling, including `AspectRation` and lowercase `roundness`, so the schema is identical from the spreadsheet to the web demo.

Feature scales differ by several orders of magnitude (Fig. 2), so LR and the MLP use a StandardScaler fitted on training inputs only; RF uses the raw values. Several predictors are almost collinear (Fig. 3): Area and ConvexArea have Pearson $r = 0.9999$, Compactness and ShapeFactor3 $r = 0.9987$, and Perimeter and EquivDiameter $r = 0.9914$. The 16 columns therefore carry far less than 16 independent pieces of information, and individual LR coefficients should not be interpreted causally.

We keep all 16 features and all nonduplicate observations, with no PCA, feature deletion, outlier removal, or skewness transformation. Because the study manipulates labels, holding preprocessing fixed across noise levels avoids introducing a second intervention.

IV. Experimental Methodology

A. Label-Randomization Mechanism

Let (x_i, y_i) be a training observation with $y_i \in \{1, \dots, K\}$ and $K = 7$. For seed s, we draw independent $u_i \sim \mathcal{U}[0, 1)$ and $z_i \sim \text{Uniform}\{1, \dots, K\}$. At randomization level p, the supplied training label is

$$\widetilde{y}_i^{(p,s)} = \begin{cases} z_i, & u_i < p, \\ y_i, & u_i \geq p. \end{cases} \tag{1}$$

This is the label-corruption scheme of Zhang et al. [1]. Because the redraw can return the original class, the expected fraction of labels that actually change is

$$\mathbb{E}[q(p, s)] = p\left(1 - \frac{1}{K}\right) = \frac{6p}{7}. \tag{2}$$

At $p = 1$ every label is redrawn but only 85.71% are expected to differ; the observed mean over three seeds is 85.93%. We record the realized q for every run.

For each seed, the same vectors u and z are reused for all $p \in \{0, 0.25, 0.50, 0.75, 1\}$, so the corrupted sets are nested as p grows, and all three models receive exactly the same label vector at a given (p, s). Labels are fixed for the whole fit and never redrawn between epochs. Validation and test labels are never changed. Equivalently, the label transition matrix is

$$T(p) = (1 - p)I + \frac{p}{K}\mathbf{1}\mathbf{1}^{\mathsf{T}}, \tag{3}$$

which is symmetric, feature-independent noise. Real annotation errors, which often depend on the class or on ambiguous inputs, are not modeled.

TABLE III
Ten clean-label candidate fits (seed 17). Selected settings are in bold.

Model	Candidate setting	Val. F1
LR	$C = 0.1$	0.9316
	$\mathbf{C = 1}$	**0.9328**
	$C = 10$	0.9320
RF	depth 8, leaf 1	0.9162
	depth 8, leaf 5	0.9172
	unlimited depth, leaf 1	**0.9239**
	unlimited depth, leaf 5	0.9217
MLP	$\alpha = 0$	0.9299
	$\alpha = 0.0001$	0.9276
	$\boldsymbol{\alpha = 0.01}$	**0.9331**

B. Models and Clean-Label Selection

Table III lists the candidate grids. LR uses an L2 penalty, LBFGS, at most 2,000 iterations, and tolerance 10^{-4}. RF uses 200 trees, bootstrap sampling, and square-root feature subsampling. The MLP has architecture $16 \to 128 \to 64 \to 7$ with ReLU activations (10,887 parameters), Adam, batch size 128, learning rate 10^{-3}, and at most 500 epochs, stopping early only if the training loss fails to improve by 10^{-5} for 20 epochs. No class weights are used.

Selection uses clean validation Macro-F1 only. Candidates within 0.001 of the best are treated as tied and resolved toward the simpler model: smaller C for LR, shallower trees and then larger leaves for RF, and larger α for the MLP. The selected settings ($C = 1$; unlimited depth with leaf size 1; $\alpha = 0.01$) were frozen before any main fit and are used at every noise level.

The main matrix contains 3 models $\times$ 5 noise levels $\times$ 3 seeds $= 45$ fits, each trained from scratch. Seeds 17, 29, and 43 control both the label noise and the estimator's random state, so the label vector is paired across models; data and model randomness are not separated.

C. Evaluation Targets and Metrics

The primary endpoint is clean-test Macro-F1 over all seven classes, with undefined per-class F1 set to zero:

$$F_{1,\text{macro}} = \frac{1}{K}\sum_{k=1}^{K} \frac{2\,\text{TP}_k}{2\,\text{TP}_k + \text{FP}_k + \text{FN}_k}. \tag{4}$$

We also report accuracy, balanced accuracy (mean class recall), per-class scores, fit time, and convergence.

On the training set we measure two different accuracies:

$$A_{\text{train,noisy}} = \frac{1}{n}\sum_i \mathbf{1}[f(x_i) = \widetilde{y}_i], \tag{5}$$

$$A_{\text{train,clean}} = \frac{1}{n}\sum_i \mathbf{1}[f(x_i) = y_i]. \tag{6}$$

The first measures agreement with the labels the model was trained on, i.e., memorization; the second measures agreement with the true labels on the same inputs. Since the first and

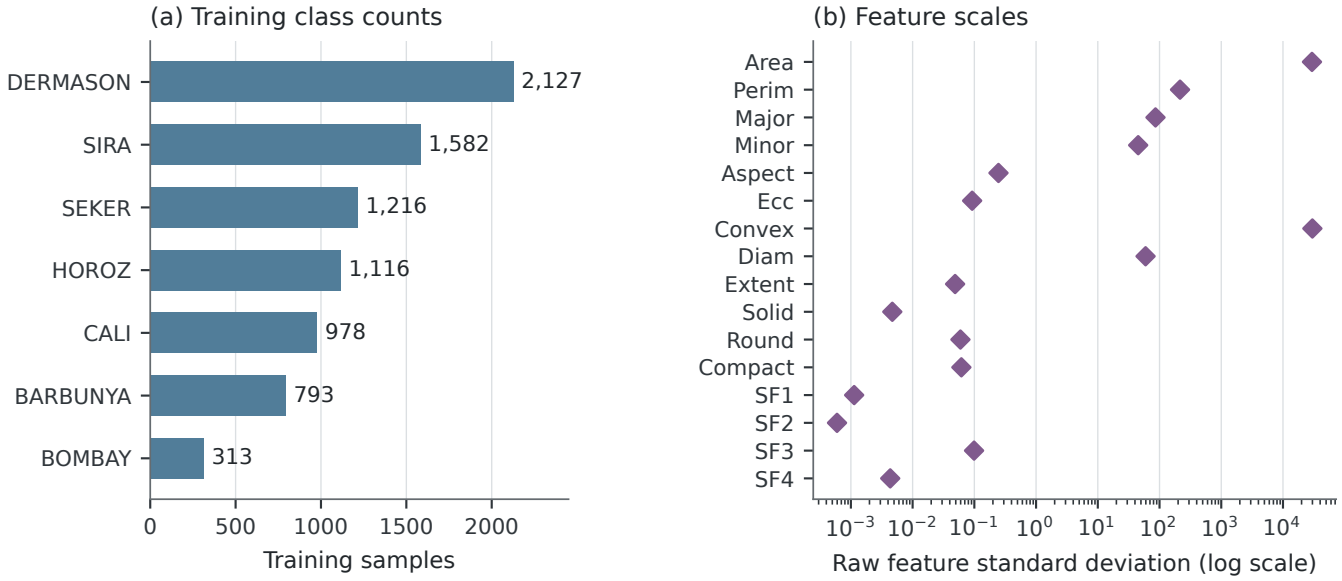


Fig. 2. Training-only exploratory analysis ($n = 8{,}125$). Left: class counts. Right: raw feature standard deviations on a logarithmic axis; the log scale is for display only and is not applied to model inputs.

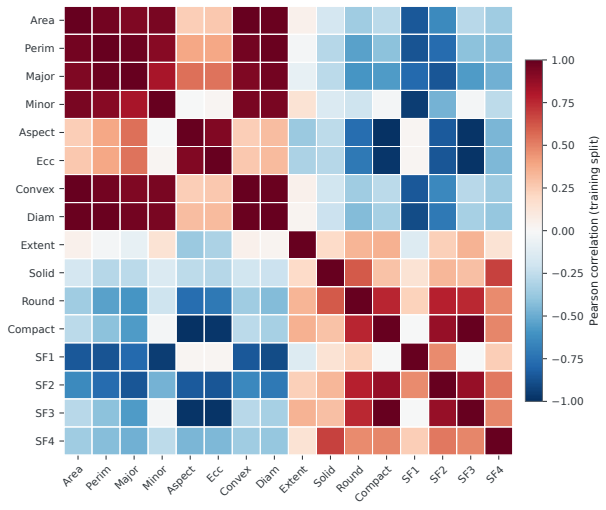


Fig. 3. Pearson correlations between the 16 features on training data. SF1–SF4 are the four shape factors; other labels abbreviate the feature names.

clean-test accuracy refer to different targets, we do not call their difference a generalization gap.

Always predicting the majority class (DERMASON) gives test accuracy 0.2617, balanced accuracy 0.1429, and Macro-F1 0.0593. Uniform random guessing has expected accuracy $1/7 \approx 0.143$.

For each model and noise level we report the mean and sample standard deviation (SD) over the three seeds. These describe run-to-run variation on one fixed split; they are not confidence intervals, and we perform no significance tests.

D. Execution Order

Tuning, the frozen configuration, and the complete main matrix were recorded before the test partition was read; an evaluation gate in the pipeline refuses to score the test set if any declared run is missing or altered. Per-observation predictions are saved so that every reported metric can be recomputed without refitting. Analyses added after the test results were known are reported separately as exploratory (Section V-E).

V. Experimental Results

A. Clean-Test Performance (RQ1)

Table IV and Fig. 4 answer RQ1. With clean training labels, test Macro-F1 is 0.9404 for LR, 0.9361 for RF, and 0.9346 for MLP, with test accuracies of 0.9280, 0.9256, and 0.9210. These small differences should not be read as a general ranking.

At $p = 0.25$, RF is the most robust (0.9280, versus 0.9164 for LR and 0.9029 for MLP). At stronger noise the order reverses: LR stays at 0.9112 at $p = 0.50$ and 0.9073 at $p = 0.75$, while RF falls to 0.8767 and 0.6440 and the MLP to 0.8391 and 0.5856. Relative to clean training, the $p = 0.75$ losses are 0.033, 0.292, and 0.349 Macro-F1 for LR, RF, and MLP. This mirrors the smooth rise of test error with corruption in Zhang et al. [1], but with strongly model-dependent slopes.

At $p = 1$ all models collapse, with Macro-F1 between 0.1144 and 0.1314 and balanced accuracy between 0.1291 and 0.1436, close to chance. LR's zero SD at $p = 0$ is expected: with identical labels and split, LBFGS converges to the same solution for every seed. At $p = 1$, LR has the largest spread (SD 0.0552), so its stability at moderate noise does not extend to complete randomization.

TABLE IV
CLEAN-TEST MACRO-F1 UNDER TRAINING-LABEL RANDOMIZATION (MEAN ± SAMPLE SD OVER THREE SEEDS). THE ACTUAL CHANGED FRACTION q IS SHARED BY ALL MODELS.

Redraw probability p	Actual changed fraction q	LR	RF	MLP
0.00	0.0000 ± 0.0000	0.9404 ± 0.0000	0.9361 ± 0.0012	0.9346 ± 0.0040
0.25	0.2156 ± 0.0052	0.9164 ± 0.0028	0.9280 ± 0.0047	0.9029 ± 0.0024
0.50	0.4306 ± 0.0071	0.9112 ± 0.0029	0.8767 ± 0.0014	0.8391 ± 0.0094
0.75	0.6486 ± 0.0078	0.9073 ± 0.0044	0.6440 ± 0.0219	0.5856 ± 0.0419
1.00	0.8593 ± 0.0058	0.1144 ± 0.0552	0.1314 ± 0.0205	0.1309 ± 0.0172

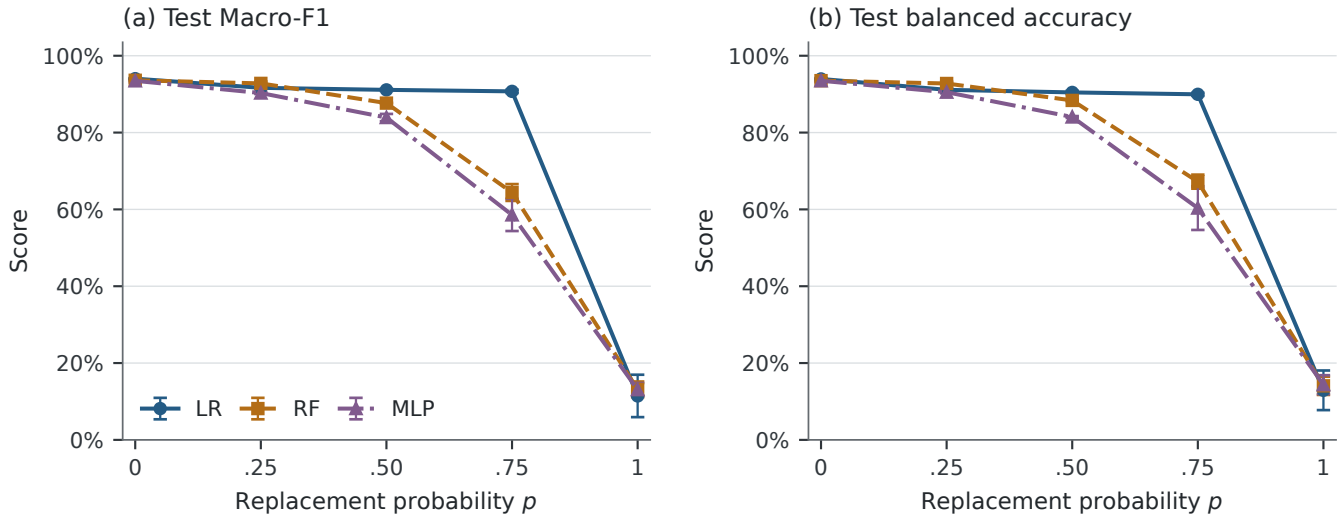


Fig. 4. Clean-test Macro-F1 (left) and balanced accuracy (right) versus training-label redraw probability. Points are three-seed means and bars are sample SD. All conditions use the same 2,709 test observations with original labels.

TABLE V
ENDPOINT BEHAVIOR AT $p = 1$ (THREE-SEED MEANS). THE TWO TRAINING COLUMNS USE THE SAME INPUTS WITH DIFFERENT LABELS. RF OOB IS THE OUT-OF-BAG ACCURACY AGAINST THE ORIGINAL TRAINING LABELS (SECTION V-E).

Model	Noisy-train acc.	Clean-train acc.	Clean-test acc.
LR	0.1602	0.1336	0.1336
RF	1.0000	0.1407	0.1378
RF OOB	0.1444	0.1391	—
MLP	0.3806	0.1432	0.1425

B. Training Agreement and Memorization (RQ2)

Figure 5 and Table V answer RQ2. RF reaches 100% accuracy on the supplied training labels at every noise level. Its accuracy on the original training labels therefore falls in step with the fraction of labels that changed: a model that reproduces every supplied answer also reproduces every injected error. At $p = 1$ its original-label training accuracy is 0.1407 and its clean-test accuracy 0.1378.

The MLP fits 38.06% of the random training labels at $p = 1$, well above LR (16.02%) but far below RF. Under its selected regularization and 500-epoch budget it memorizes only partially, and its clean-test Macro-F1 is as low as RF's. In both cases, agreement with randomized supervision says nothing about clean-test quality.

At $p = 0.75$, RF still fits the supplied labels perfectly, but its clean-test accuracy (0.6652) is far above its original-label training accuracy (0.3514). The training predictions copy a heavily corrupted label vector, while predictions on new points average over many trees and neighbors and can still exploit the structure carried by the 35% of labels that were not changed.

C. Class-Level Behavior

Figure 6 shows row-normalized test confusion matrices at both endpoints, pooled over the three seeds. BOMBAY, the smallest class, is perfectly separated with clean labels: LR and RF recall all 104 BOMBAY test observations, so small class size alone does not make a class hard. SIRA is the hardest class; for LR its clean-label F1 is 0.8687, while BOMBAY reaches 1.0000. Under complete randomization the diagonal disappears, and the off-diagonal pattern differs by model and seed rather than being uniform.

D. Optimization and Computation

Figure 7 relates the MLP results to its finite budget. Fourteen of the 15 main MLP runs reach the 500-epoch cap with a convergence warning; only the clean-label run with seed 17

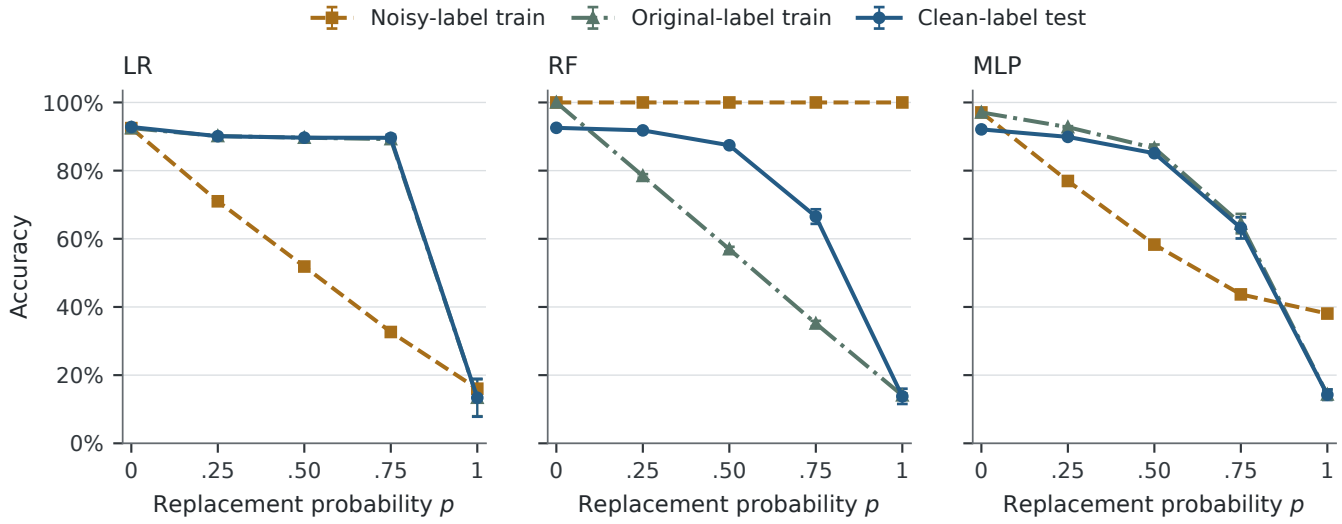


Fig. 5. Accuracy against three targets for each model: supplied (possibly randomized) training labels, original training labels, and original test labels. Means and sample SD over three seeds. RF fits the supplied labels perfectly at every noise level while its clean-test accuracy deteriorates.

stops earlier, at 430 epochs. The training loss decreases far more slowly as p grows (at $p = 1$ it falls only from 1.97 to 1.66 in 500 epochs), consistent with the slower convergence on random labels reported by Zhang et al. [1]. Because the curves optimize different targets, their levels do not compare clean-test quality.

Fitting took 462.8 s in total for the 55 tuning and main fits (401.0 s for the main matrix: 2.6 s LR, 33.2 s RF, 365.3 s MLP) on a 20-thread Windows 11 CPU machine with BLAS restricted to one thread and four RF workers.

E. Exploratory Follow-up

The analyses in this subsection were run after the test results were known and are therefore exploratory. They use only training and validation data, and they do not change any result above.

RF out-of-bag accuracy. We refitted the 15 main RF conditions with `oob_score=True`, which does not change how trees are grown; every refit reproduced the recorded clean-test Macro-F1 exactly. Out-of-bag (OOB) accuracy against the original training labels is 0.9216, 0.9150, 0.8644, 0.6357, and 0.1391 for $p = 0$ to 1, closely tracking clean-test accuracy (0.9256, 0.9181, 0.8744, 0.6652, 0.1378). OOB accuracy against the supplied labels at $p = 1$ is 0.1444, at chance, while in-bag accuracy is 1.0000. The forest's memorization is thus visible from the training data alone, provided predictions are scored on points each tree did not see.

MLP capacity. The main MLP fits only 38% of random labels, which raises the question of whether this is a limit of the model or of the budget. We ran the capacity study specified in our protocol before execution: a fixed stratified subset of 3,000 training rows, widths (32), $(128, 64)$, and $(512, 256)$, $p \in \{0, 1\}$, the same three seeds, $\alpha = 0$, and exactly 1,000 epochs (18 fits). Table VI reports the results. Capacity has a large effect. On random labels, the 775-parameter network fits 24.4% of the subset, the main 128–64 architecture without weight decay fits 63.7%, and the 141,831-parameter 512–256 network fits 96.6% (98.1% for seed 17), with its training loss still trending downward at the last epoch. Trained on clean labels, all three widths reach validation Macro-F1 between 0.927 and 0.932, and on random labels all fall to 0.14–0.16. As in Zhang et al., the network that can almost memorize random labels also fits clean labels perfectly and still generalizes well on them, so its capacity alone does not predict how well it generalizes. Each 512–256 fit took about 150 s on a four-core Linux machine with the same package versions.

TABLE VI
EXPLORATORY MLP CAPACITY STUDY ON A 3,000-ROW TRAINING SUBSET ($\alpha = 0$, 1,000 EPOCHS; MEAN $\pm$ SAMPLE SD OVER THREE SEEDS). VALIDATION USES THE CLEAN 2,709-ROW VALIDATION SET.

Hidden layers	Params	p	Noisy-train acc. / Val. F1
(32)	775	0	0.955 ± 0.001 / 0.932 ± 0.001
		1	0.244 ± 0.008 / 0.159 ± 0.006
(128, 64)	10,887	0	0.997 ± 0.001 / 0.927 ± 0.005
		1	0.637 ± 0.025 / 0.140 ± 0.042
(512, 256)	141,831	0	1.000 ± 0.000 / 0.929 ± 0.001
		1	0.966 ± 0.013 / 0.140 ± 0.019

VI. Discussion

A. Why Partial Noise Can Preserve Signal

LR's strong result at $p = 0.75$ has a simple population-level explanation. Under the uniform replacement of Eq. (3), let $\eta_k(x) = \Pr(Y = k \mid X = x)$ be the clean class posterior. The corrupted posterior is

$$\widetilde{\eta}_k(x) = (1 - p)\eta_k(x) + \frac{p}{K}, \tag{7}$$

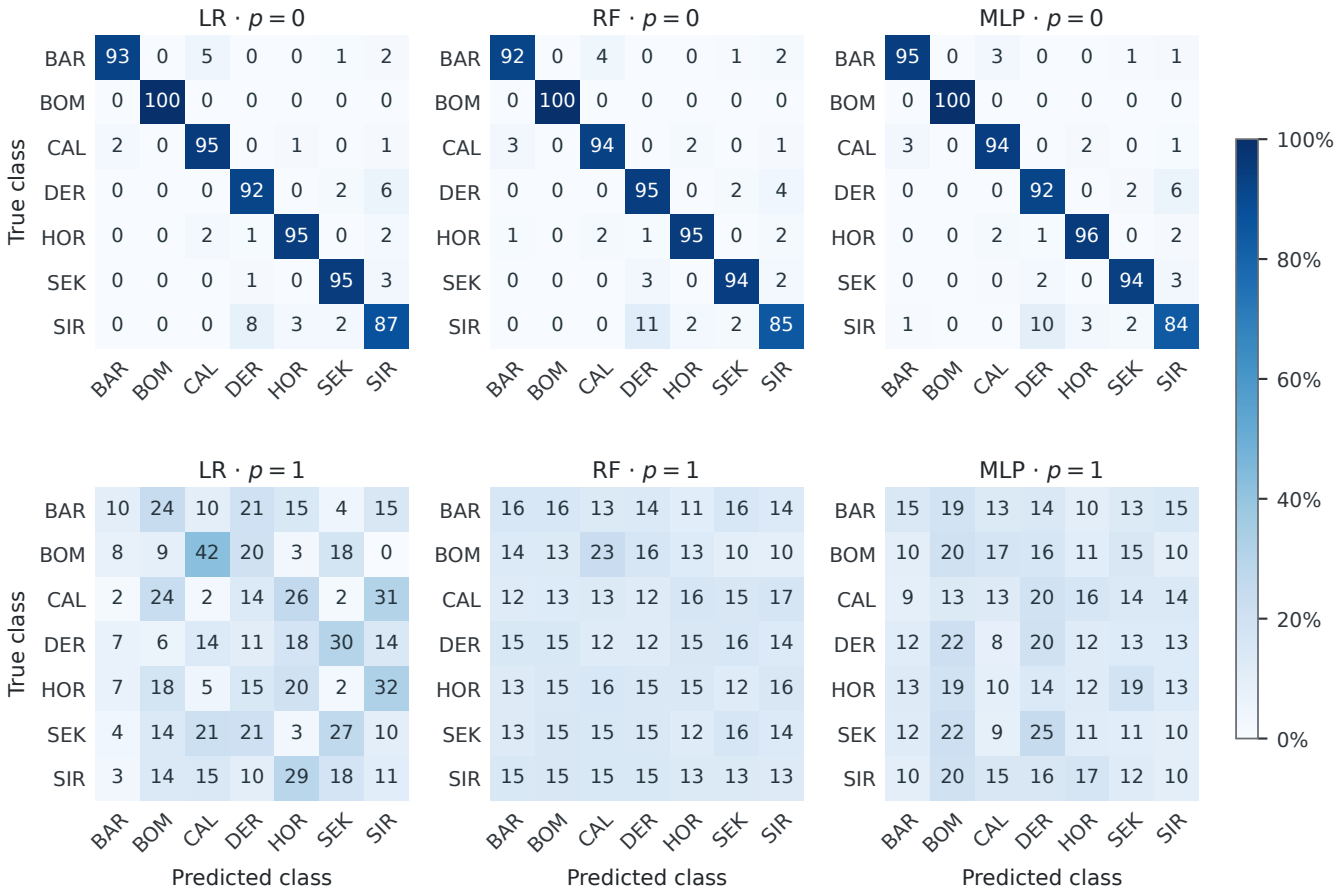


Fig. 6. Row-normalized test confusion matrices for LR, RF, and MLP (columns) after clean ($p = 0$, top) and fully random ($p = 1$, bottom) training labels. Counts are summed over three seeds before normalizing.

so for any two classes

$$\widetilde{\eta}_a(x) - \widetilde{\eta}_b(x) = (1-p)\big(\eta_a(x) - \eta_b(x)\big). \tag{8}$$

For $p < 1$ the ranking of classes at every x is preserved and only the margins shrink by $1-p$, so the Bayes-optimal classifier is unchanged; this is the classical observation behind noise-robust learning under symmetric noise [7], [8], [9]. At $p = 1$ all posteriors equal $1/K$ and no label information remains.

A model that estimates smooth posteriors from many points, such as regularized LR, benefits directly: noisy labels average out, and the decision boundary stays close to the clean one. Models that can isolate individual points, such as a fully grown RF or a flexible MLP, instead fit many corrupted labels locally, and those local errors transfer to nearby test points once the noise is dense enough. Finite samples, model misspecification, and the fact that Macro-F1 is not the zero–one loss all limit how far this argument goes, so it explains the trend rather than proving LR robustness in general.

B. Connection to Zhang et al.

RF reproduces the core observation of Zhang et al. on tabular data: a model can fit fully random labels perfectly while its predictions on new data are at chance. Our main MLP does not, but the capacity study suggests the difference is one of capacity, regularization, and training budget rather than of model family. Like the original paper, we find that explicit regularization alone does not determine whether random labels can be fitted, and that the effective capacity of a trained model depends on the optimizer and budget as much as on the architecture. The RF OOB result adds a practical point: for bagged models, a standard internal estimate already detects memorization without any held-out data.

C. Threats to Validity

One dataset and one split. Seeds vary label noise and model initialization on a single split. Results may differ on other splits or datasets, and repeated test predictions on the same 2,709 observations are not independent samples.

Fixed clean-label hyperparameters. Tuning on clean labels and freezing the choice makes the intervention interpretable, but it can disadvantage a model that would benefit from stronger regularization under noise. The small grids do not equalize effective capacity across families.

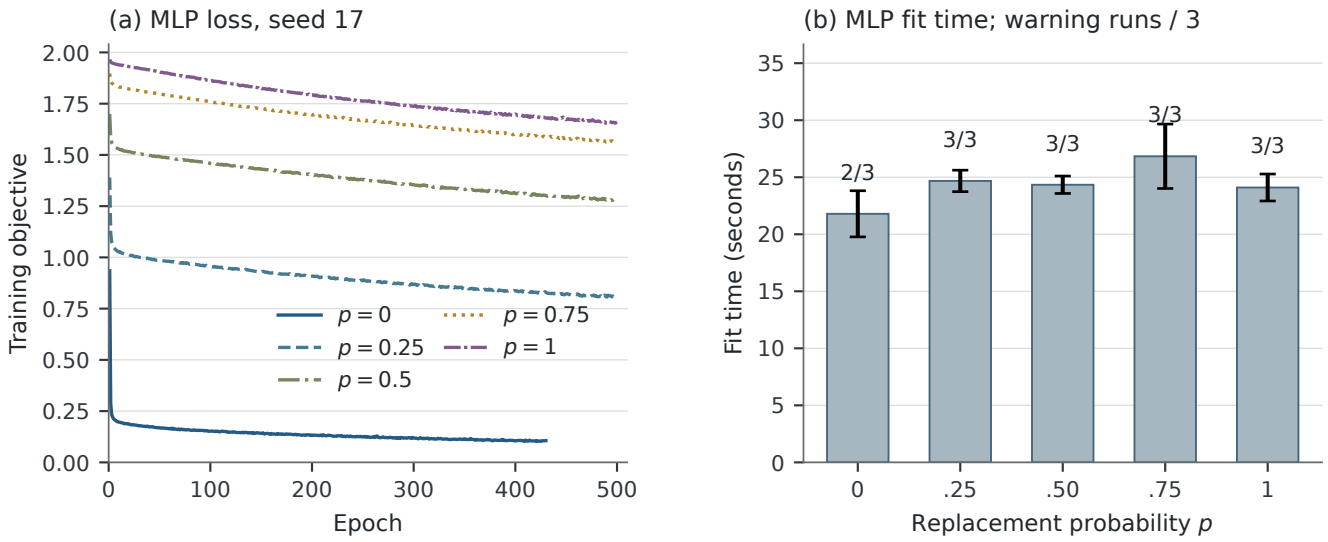


Fig. 7. MLP optimization. Left: training loss for seed 17 at the five noise levels; each curve optimizes a different label vector. Right: fit time over the 15 main MLP runs (mean ± sample SD), annotated with the number of runs per condition that reached the epoch cap.

Optimization budget. Most main MLP fits reach the epoch cap, so capacity, regularization, and optimization effects are confounded in the main matrix; the capacity study addresses this only on a subset and after the holdout was seen.

Synthetic noise. Uniform, feature-independent replacement is the most benign kind of label noise. Class-dependent or boundary-concentrated errors would not preserve posterior ordering, and our conclusions may not transfer to them.

VII. Reproducibility and Web Demo

The implementation uses Python 3.12, scikit-learn 1.7.2, NumPy 2.2.6, pandas 2.2.3, SciPy 1.15.3, and Matplotlib 3.10.7, with all transitive versions locked. A staged pipeline covers data preparation, EDA, tuning, training, evaluation, reporting, and an automated test suite. Rerunning the complete pipeline in a fresh directory and environment on the original machine reproduced every non-runtime value in the tuning and main tables exactly. A rerun on Linux selected the same parameters and matched 40 of the 45 main runs exactly; the remaining iterative fits differed by at most 0.0006 in any condition mean. Trained models and row-level predictions are regenerated by the pipeline rather than stored in the repository; the paper figures can be rebuilt from the recorded tables without fitting any model.

A Flask web demo, deployed at https://ecs171-label-demo.vercel.app, loads six saved pipelines: one seed-17 model per family trained with clean ($p = 0$) and fully random ($p = 1$) labels. Users enter or load the 16 measurements of a validation bean and compare the two models' seven-class probabilities. Preprocessing is stored inside each pipeline, so the demo cannot apply different scaling statistics. Inputs are validated, values outside the training range trigger a notice instead of being clipped, and the endpoint performs inference only. On all seven example beans, the six deployed models matched offline predictions, with a maximum probability difference of 2.8×10^{-17}. Displayed probabilities are uncalibrated model outputs.

VIII. Conclusion

Label randomization cleanly separates reproducing training answers from predicting true labels on new data. On Dry Bean, a random forest fits fully random labels perfectly while its clean-test Macro-F1 falls to 0.1314, and its out-of-bag accuracy reveals the collapse from the training set alone. The MLP memorizes only partially under its original budget, but a wider unregularized network trained longer fits 97% of random labels on a training subset, approaching the behavior reported by Zhang et al. Under heavy partial noise, regularized logistic regression keeps most of its clean performance, consistent with the preserved posterior ordering of symmetric noise, while the more flexible models degrade quickly.

The main lesson is methodological: always state which labels define each metric, record the exact corruption mechanism, and judge models on unseen observations with the original labels. Similar clean-label scores can hide very different responses to corrupted supervision. Natural next steps are class-dependent noise, the shuffled- and random-input controls of the original paper, and repeating the study over multiple splits and tabular datasets.

IX. Project Roadmap, Contributions, and Availability

A. Roadmap

Table VII summarizes the project milestones.

TABLE VII
PROJECT MILESTONES.

Milestone	Outcome
Problem definition	Assigned paper reviewed; RQ1/RQ2 and noise protocol specified.
Data and EDA	Source audit, deduplication, fixed split, training-only EDA.
Model development	Ten clean-label tuning fits; settings frozen.
Main experiments	45 fits with saved predictions and training records.
Evaluation	Single clean-test evaluation; 15 condition summaries.
Exploratory follow-up	RF out-of-bag analysis; 18-fit MLP capacity study.
Web demo	Public deployment with verified inference.
Proposal (Oct. 21)	Submitted.
Mid-quarter report	Submitted.
Final report and repository	Completed.
Presentation (Nov. 11)	Five-minute talk with live demo.

B. Team Contributions

Linjie Yang led the implementation, experiments, analysis, reproducibility tooling, web demo, and drafting of the report. Tianying Chen coordinated the team, tested the web interface, and organized the deliverables and submission. Elsey Chen checked the dataset and references and independently reviewed the figures and reported metrics. All authors reviewed the final report and presentation.

C. Code and Data Availability

All code, configuration, recorded results, and the LaTeX source of this report are publicly available at

https://github.com/Felix-Yang-369/ecs171-label-randomization.

The original code is released under the Apache License 2.0. The Dry Bean data are redistributed unmodified under CC BY 4.0, and the IEEE class files retain their own LPPL license. The README and `REPRODUCIBILITY.md` give setup instructions for Windows, macOS, and Linux.

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